Demonstration of Proposed Features

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimisation Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the OptiCrop project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the project's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	**Crop Recommendation**	Accepts N, P, K, temperature, humidity, pH, and rainfall as inputs and returns the most suitable crop using the trained Random Forest classifier (99.55% test accuracy).	**Implemented**	**Yes**	Successfully demonstrated on the Find Your Crop page — entering rice-profile values returns 'Rice' correctly.
2	**Multi-Algorithm Model Comparison**	Trains and compares four supervised classifiers — Logistic Regression, KNN, Decision Tree, and Random Forest — on identical train/test splits. Best model selected based on accuracy.	**Implemented**	**Yes**	Demonstrated in the Jupyter notebook — model comparison table shows Random Forest at 99.55% accuracy.
3	**K-Means Clustering Analysis**	Applies K-Means clustering with the Elbow Method (k=4) to discover natural agro-climatic groupings in the soil and climate data for research and policy insights.	**Implemented**	**Yes**	Demonstrated in the notebook — elbow plot and cluster groupings shown successfully.
4	**User Input Form**	A clean, responsive web form on the Find Your Crop page that collects all seven soil and climate parameters from the user with clearly labelled input fields and a Predict button.	**Implemented**	**Yes**	Demonstrated live — form accepts all 7 inputs and renders the prediction result

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					on the same page.
5	**Input Validation & Error Handling**	Validates all seven form fields on submission. Non-numeric or missing inputs are caught by a try/except block and return a plain error message without crashing the server.	**Implemented**	**Yes**	Demonstrated — empty form submission returns a clean error message, application remains stable.
6	**Three-Page Flask Web Application**	Home page (introduction and navigation), About page (project background and model performance), and Find Your Crop page (prediction form) — all accessible from a Bootstrap 5 navigation bar.	**Implemented**	**Yes**	All three pages load correctly and navigation between them works smoothly.
7	**GitHub Repository & Version Control**	Complete project hosted at github.com/neelimavana/OptiCrop with all source code, dataset, trained model artifacts, Jupyter notebook, requirements.txt, and README.	**Implemented**	**Yes**	Repository is public and accessible — mentor can clone and run the application directly.
8	**User Authentication**	Secure login and registration system for farmers to access personalized crop recommendations.	**Pending**	**No**	Planned for future scope — requires database integration (SQLite/MySQL) and Flask session management.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	8
Total Features Implemented	7
Total Features Demonstrated	7
Features Pending (Future Scope)	1
Overall Implementation Rate (%)	87.5%

Note:

Feature 8 (User Authentication) was identified during project planning based on the ER diagram (which includes a User entity with user_id, name, email, password). It was not implemented in the current version due to the 7-day project timeline and the requirement to deliver a working ML pipeline and web application first. It is documented as planned future scope requiring database integration.